

Analysis of IPv4 Header Problem Solving and Protocol Fundamentals

Overview

The IPv4 header functions as the control core of the Network Layer. Effective problem solving requires:

- Understanding field scaling rules (e.g., $HLEN \times 4$)
- Correct hexadecimal-to-decimal conversion
- Knowing which fields routers modify
- Applying encapsulation logic across layers

This document consolidates mathematical constraints, router behavior, and layered interactions relevant to competitive exams and technical analysis.

1. Field Calculations and Header Constraints

1.1 Header Length (HLEN)

- Size: 4 bits
- Represents number of 32-bit words
- Actual header size = $HLEN \times 4$ bytes

Valid Range

HLEN Value	Header Size
5	20 bytes (minimum)
15	60 bytes (maximum)

Key Rules

- Must always be a multiple of 4 bytes.
- If header size is not multiple of 4 → padding is added.

- If $HLEN < 5 \rightarrow$ packet is invalid and discarded.

1.2 Total Length Field

- Size: 16 bits
- Represents entire datagram size (Header + Data)
- Maximum value: 65,535 bytes

Data Size Formula

Data Size = Total Length - ($HLEN \times 4$)

Hexadecimal Conversion Example

If Total Length = 0084 (hex):

$$0084_{16} = 132_{10}$$

If $HLEN = 5$:

$$\text{Header} = 5 \times 4 = 20 \text{ bytes}$$

$$\text{Data} = 132 - 20 = 112 \text{ bytes}$$

2. Time to Live (TTL) and Loop Prevention

2.1 Field Properties

- Size: 8 bits
- Range: 1–255 at source
- 0 not used for transmission

TTL limits hop count, not time in seconds.

2.2 Router Decrement Rule

- Each router decrements TTL by exactly 1.
- TTL does NOT decrement inside a LAN.
- If TTL becomes 0 at router → packet discarded.
- ICMP TTL Exceeded sent to source.

If TTL becomes 0 exactly at destination → packet accepted.

2.3 Hop Analysis Example

Initial TTL = 32

If packet crosses 6 routers:

Final TTL = $32 - 6 = 26$

This matches common competitive exam scenarios.

3. Router Modifications and Field Mutability

Understanding which fields change is critical.

3.1 Fields That Always Change

- TTL (decremented at each hop)
- Header Checksum (recalculated at each hop)

Checksum must be recalculated because TTL changes.

3.2 Fields That Change Conditionally

- Total Length (during fragmentation)
- Fragment Offset
- MF flag
- Header Length (if options present)

3.3 Fields That Never Change

- Version
- Source IP address
- Destination IP address
- Protocol field
- Identification (remains same across fragments)
- DF flag (unless set by source)

3.4 Common Misconception

Port numbers are NOT part of the IP header.

They belong to:

- TCP header
- UDP header

IP only identifies the protocol number (e.g., 6 for TCP, 17 for UDP).

4. Encapsulation and Payload Size Analysis

When a TCP segment is encapsulated inside an IPv4 datagram:

From IP's perspective:

Data = Entire TCP Segment

TCP Segment = TCP Header + TCP Data

4.1 General Formula

TCP Data Size =
Total Length (IP)
– IP Header Size
– TCP Header Size

4.2 Example

Given:

Total IP Length = 1000 bytes

IP HLEN = 10 $\rightarrow 10 \times 4 = 40$ bytes

TCP HLEN = 15 $\rightarrow 15 \times 4 = 60$ bytes

Application Data:

$1000 - 40 - 60 = 900$ bytes

5. Error Detection Across Layers

Different layers use different mechanisms.

Layer	Mechanism
Data Link (Ethernet)	CRC
Network (IP)	Header Checksum
Transport (TCP/UDP)	Checksum

Important distinction:

IP checksum covers header only.

TCP checksum covers header + data (pseudo-header included).

6. Options Field Constraints

- Maximum options size = 40 bytes
- Total maximum header = 60 bytes

Record Route Limitation

Options = 40 bytes

2 bytes used for Type + Length

Remaining = 38 bytes

Each IP address = 4 bytes

Maximum routers recorded ≈ 9

7. Efficiency Calculations

Network efficiency:

Efficiency = Data / Total Length

Example:

Data = 140 bytes

Total Length = 160 bytes

Efficiency = $140 / 160 = 87.5\%$

8. Problem-Solving Strategy Summary

For competitive exams and technical analysis:

Step 1: Identify Given Format

- Binary?
- Hexadecimal?
- Decimal?

Step 2: Convert Units

- Hex $\rightarrow$ Decimal
- HLEN $\times$ 4
- TCP HLEN $\times$ 4

Step 3: Apply Correct Formula

- Data = Total - Header
- TCP Data = IP Total - IP Header - TCP Header
- Final TTL = Initial TTL - Number of routers

Step 4: Validate Constraints

- HLEN $\geq$ 5?
- Total Length $\geq$ Header?
- TTL $>$ 0?